

2026-06-10-impl-batch1-review

Code-Review Report — Impl Batch 1 (SP1 / SP2 / SP3)

Reviewer: Independent code-review subagent (§11.4.125 / §11.4.142) **Date:** 2026-06-10 **Scope:** Uncommitted working-tree batch (D-1..D-4, D-6, D-9, CAF scripts) across `helix_code/` (SP1), `submodules/helix_agent/` (SP2), `scripts/caf/` (SP3). **Method:** Read every diff; ran real read-only builds, vet, RED/GREEN polarity, pin guards, the SP3 Challenge, and a paired §1.1 mutation. No edits/commits/pushes (mutation residue restored).

VERDICT: GO

All builds + vet clean, all GREEN guards pass, every §11.4.115 polarity / pin / mutation behaves correctly (FAILs only where it must), the SP3 Challenge is 28 PASS / +1 FAIL under `--mutate`, the §11.4.120 reconciled tests are re-pointed (not weakened to tautologies), no `go.mod/` cross-area drift, and no `simulated|for now|TODO` implement introduced in production source. The two observed test FAILs are pre-existing intermittent flakes provably unrelated to the batch.

- **BLOCKER findings:** 0
 - **Must-fix-before-merge findings:** 0
 - **Non-blocking notes:** 2 (below)
-

Captured evidence (commands run this session)

1. Builds + vet

```
helix_code: go build ./internal/verifier/... ./internal/secrets/... ./c
helix_code: go vet   ./internal/verifier/... ./internal/secrets/... ./c
helix_agent: go build ./internal/handlers/... ./internal/router/... ./in
helix_agent: go vet   ./internal/handlers/... ./internal/router/... ./in
```

2. SP1 GREEN suite (RED_MODE=0 — all PASS)

```
ok dev.helix.code/internal/verifier 4.768s
ok dev.helix.code/internal/secrets  1.033s
ok dev.helix.code/cmd/cli            2.134s
```

3. Polarity / RED behaviour (§11.4.115)

```
SP2 D1 RED_MODE=1 TestCompletionModels_D1_Polarity      → FAIL (defec
SP1 secrets RED_MODE=1 TestLoadAPIKeys_GapFillPrecedence → FAIL ("F00=
SP1 verifier/cli RED_MODE=1                             → PASS (see N
SP2 handlers GREEN (default) Models tests               → PASS
```

4. Pin guards (PIN_STUB_BLUFF=1 — must FAIL; skip by default)

```
D6 qwencode    TestD6_QwenCode_GenerateIsStubBluff          → FAIL ( "// Ge
D9 instmgr     TestD9_InstanceManager_ExecuteStubsAreBluffs → FAIL ( aider/
Default run (no env):                                     both SKIP (
```

5. Full SP2 affected packages (default mode)

```
ok dev.helix.agent/internal/handlers / clis/** / router    (all ok except t
```

6. SP3 Challenge

```
bash scripts/caf/challenge_caf.sh          → === RESULT: 28 PASS, 0 FAIL
bash scripts/caf/challenge_caf.sh --mutate  → === RESULT: 28 PASS, 1 FAIL
```

7. Paired §1.1 mutation on the new code (anti-tautology proof)

```
Mutate GetWorkingModels score-gate to no-op ( _ = minScore) →
TestGetWorkingModels_VerifiedAndScore FAIL: "sub-threshold model must be
Restore → ok dev.helix.code/internal/verifier 0.650s ; residue markers = 0
```

Soundness assessment

- **D-2/D-4 (working-model funnel):** GetWorkingModels correctly ANDs key-presence $\wedge$ Verified $\wedge$ status=="verified" $\wedge$ score>=GetMinAcceptableScore(). Mutation proved the score-gate is load-bearing. printVerifiedModels defensively drops non-working rows; Priority-3 fallback now uses dedicated printFallbackModels with an explicit “ Δ unverified fallback” label — no path renders an unverified model as verified. Priority-2 uses its own cli_model_info_provider rendering and is unaffected by the stricter printer.
- **D-3 (secrets gap-fill):** setIfAbsent honours DECISION-1 (shell-exported wins; empty treated as gap; values not logged — CONST-042). Wired into main() before any config read. RED_MODE=1 on the LIVE path correctly FAILs (fix present).
- **D-1 (helix_agent Models):** hardcoded 3-model literal removed; list now sourced from the live ProviderRegistry (ListProviders→GetProvider→GetCapabilities().SupportedModels), honest-empty when no source. SetModelSource(providerRegistry) wired at router.go:1196. The D1 polarity test drives the REAL handler and proves the list tracks registry state — true live-artifact RED/GREEN.
- **§11.4.120 reconciliation:** the three reconciled tests (_Direct, _ResponseStructure, Unit_Models) are re-pointed to assert the NEW mechanism (honest-empty with no source + registry-tracked with a staticModelSource) and explicitly assert the removed literals are NOT fabricated. They are NOT deleted and NOT weakened to tautologies.
- **D-6/D-9 pin guards:** genuine — FAIL under PIN_STUB_BLUFF=1 against the live stubs, skip with SKIP-OK by default. Correctly left RED to pin BLUFF-001/003 for SP4.
- **Anti-bluff source scan:** no simulated|for now|TODO implement|in production this would introduced in any production source line of the batch.

Scope discipline

- No `go.mod/go.sum` edits in either repo.
- SP1 confined to `helix_code/{cmd/cli/main.go, internal/secrets/loader.go, internal/verifier/adaptor.go}` + its 4 new `*_test.go`.
- SP2 confined to `submodules/helix_agent/internal/{handlers,router,clis}` per its declared lane.
- SP3 confined to `scripts/caf/`.
- No cross-area edits observed.

Non-blocking notes

- **Note A (SP1 RED rigor):** the SP1 verifier (`TestGetWorkingModels_VerifiedAndScore`) and `cli` (`TestD2_PrinterDropsNonWorkingRows`) RED-mode branches assert against a *captured copy* of the old logic (raw `GetVerifiedModels`, `oldDisplayPredicate`) rather than reverting the live artifact, so `RED_MODE=1` PASSES for them. The GREEN branches DO exercise the real new functions and the §1.1 mutation proved they are not tautologies, so anti-bluff is satisfied — but these are weaker than the secrets/SP2 live-artifact RED pattern. Optional future hardening, not a blocker.
- **Note B (pre-existing flakes — confirmed out-of-scope):** `TestCompletionHandler_IDGeneration` (`completion_extended_test.go:580`) and `TestGenerateTeamID` (`ensemble_handler_teams_test.go:496`) flake on timestamp-collision uniqueness asserts. The batch touched NEITHER the test bodies NOR the functions under test (`convertToInternalRequest`, `generateTeamID`) — `git diff` confirms no change to them. In isolation they failed 2/5 runs and passed 3/5, proving intermittent pre-existing flakiness, NOT a regression introduced by this batch.

Final verdict line

GO — BLOCKERS: 0 | must-fix: 0 | non-blocking notes: 2. Observed PASS/FAIL: - builds/vet: BUILD_EXIT=0 (both repos), VET_EXIT=0 (both repos) - SP1 GREEN: ok `internal/verifier` 4.768s / ok `internal/secrets` 1.033s / ok `cmd/cli` 2.134s - Polarity FAIL-where-required: SP2 D1 `RED_MODE=1` FAIL ✓ ; SP1 secrets `RED_MODE=1` FAIL ✓ - Pin guards FAIL-where-required: D6 FAIL ✓ ; D9 FAIL ✓ (skip by default ✓) - SP3 Challenge: 28 PASS, 0 FAIL ; --mutate → 28 PASS, 1 FAIL ✓ - §1.1 mutation on `GetWorkingModels`: guard FAILs under mutation, PASSES restored ✓ (residue=0) - Pre-existing flakes (out-of-scope): `IDGeneration` / `GenerateTeamID` — intermittent, untouched by batch